

Leyi (Michelle) Huang

San Diego, CA | michellely.huang728@gmail.com | 858-539-6364 | linkedin.com/in/leyi-huang-michelle | https://github.com/michellelyHuang728

EDUCATION

University of California, San Diego (UCSD) | San Diego, CA

Sep 2025 – Present

Master of Science in Computer Science

Relevant Coursework: Parallel & High-Performance Computing, Database Systems, Statistical NLP, Computer Security, Image Synthesis

University of California, San Diego (UCSD) | San Diego, CA

Sep 2021 – Jun 2025

Bachelor of Science in Mathematics–Computer Science | GPA: 3.768 | Provost Honors

Relevant Coursework: Software Engineering, Operating Systems, Data Structures & OOP, Data Science, Machine Learning

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), Python, C/C++, Rust, SQL, HTML5, CSS3, Bash, Assembly

Frontend: Node.js, React, WebDriver.io, Responsive Design

Backend & APIs: Node.js/Express, FastAPI, Java, Python, PyTorch, TensorFlow, REST API design, Numpy, Pandas, SQL, PostgreSQL

Tools & Platforms: Docker, Kubernetes, AWS, Linux, Git, GitHub Actions (CI/CD), JUnit, CUDA C++, Unreal Engine 5

EXPERIENCE

Knight Lab @ UC San Diego | San Diego, CA

Aug 2026 – Present

Graduate Research Assistant – High-Performance Computing

- Engineered an **out-of-core PCoA algorithm in Rust/C++** leveraging randomized sketching and multi-pass matrix algorithms, eliminating memory overhead by avoiding full matrix materialization.
- Implemented and benchmarked **CPU-accelerated numerical kernels** with **BLAS/OpenMP**, targeting scikit-bio compatibility for large-scale datasets while ensuring strict memory safety and computational efficiency.

Qualcomm Institute (Calit2) | San Diego, CA

Sep 2024 – Present

Software Engineer Intern – Serious Game Team

- Architected a **modular C++/Unreal Engine 5 simulation engine**, decomposing complex biological processes into independently testable subsystems to reduce feature integration time by ~35% across a 3-person engineering team.
- Optimized **3D rendering pipeline** via LOD tuning and object pooling (cutting GPU memory by ~25%) for K-12 education platforms.

System, Energy and Efficiency Lab @ UCSD | San Diego, CA

Apr 2024 – Jun 2025

Undergraduate Research Assistant

- Quantized CasaNovo and Llama models using **INT8/FP16 techniques**, reducing model size by ~4x to deploy **containerized (Docker) inference** on Raspberry Pi 5 within a 200 MB memory budget.
- Engineered automated reliability monitoring script in Python to capture inference latency, throughput, and memory usage across 10+ benchmark runs, surfacing bottlenecks that cut average inference time by ~30%.

Perfect World Games | Beijing, China

Jul 2023 – Sep 2023

Software Engineer Intern – Algorithm Team

- Architected scalable **Python backend REST APIs** to fine-tune and serve **multilingual NLP models** (text2vec, multilingual-e5-base, BAAI/bge), processing 10,000+ Chinese text pairs and achieving >90% semantic similarity classification accuracy.
- Developed a **text-to-speech (TTS) synthesis API** in Python (PyTorch/TensorFlow) integrating a **fine-tuned VITS** model, supporting English, Japanese, and Chinese voice generation across 500+ character voice profiles.
- Automated model evaluation and **benchmarking** model performance across fine-tuning iterations within **CI/CD workflows**, reducing manual evaluation cycles by ~30% and accelerating model delivery by 1 sprint.

PROJECTS

American Gut Project Knowledge Platform | San Diego, CA

July 2026 – Present

Data Engineering | Generative AI, RAG, Python, PyTorch, FAISS, LLMs

- Built an **Agentic RAG system** over 13K+ PubMed papers with **dense retrieval**, reranking, **LLM tool calling** for grounded QA.
- Designed an end-to-end pipeline for semantic chunking, embedding, and **FAISS vector indexing** with source-level citation tracing.
- Developed **FastAPI backend microservices** for local **LLM inference and RAG evaluation**, eliminating proprietary API dependencies.

CPU/GPU-Accelerated Matrix Multiplication |

Sep 2025 – Present

Performance Engineering | CUDA C++, SIMD, AWS (g4dn.xlarge)

- Implemented **high-performance CUDA C++** and **CPU SIMD kernels** on **AWS** (NVIDIA T4), achieving ~4,000 GFLOPS (GPU via shared memory tiling) and ~21 GFLOPS (CPU SIMD) peak throughput with automated benchmarking suites.

Task List & Journal Calendar |

Mar 2024 – Jun 2024

Full-Stack Web Application | HTML, CSS, Node.js, WebDriver.io, GitHub Actions

- Developed a **full-stack web application** (Node.js/JavaScript/HTML/CSS) featuring 100% test coverage via WebDriver.io/JUnit and automated **GitHub Actions CI/CD** pipelines, cutting QA overhead by ~60%.